

# Overview of Sentiment analysis

*Subject: Sentiment analysis*

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## 1.

**Feature/aspect-based** It refers to determining the opinions or sentiments expressed on different features or aspects of entities, e.g., of a cell phone, a digital camera, or a bank. A feature or aspect is an attribute or component of an entity, e.g., the screen of a cell phone, the service for a restaurant, or the picture quality of a camera. The advantage of feature-based sentiment analysis is the possibility to capture nuances about objects of interest. Different features can generate different sentiment responses, for example a hotel can have a convenient location, but mediocre food. This problem involves several sub-problems, e.g., identifying relevant entities, extracting their features/aspects, and determining whether an opinion expressed on each feature/aspect is positive, negative or neutral. The automatic identification of features can be performed with syntactic methods, with topic modeling, or with deep learning. More detailed discussions about this level of sentiment analysis can be found in Liu's work.

## 2.

**Types** A basic task in sentiment analysis is classifying the polarity of a given text at the document, sentence, or feature/aspect level—whether the expressed opinion in a document, a sentence or an entity feature/aspect is positive, negative, or neutral. Advanced, "beyond polarity" sentiment classification looks, for instance, at emotional states such as enjoyment, anger, disgust, sadness, fear, and surprise. Precursors to sentimental analysis include the General Inquirer, which provided hints toward quantifying patterns in text and, separately, psychological research that examined a person's psychological state based on analysis of their verbal behavior. Subsequently, the method described in a patent by Volcani and Fogel, looked specifically at sentiment and identified individual words and phrases in text with respect to different emotional scales. A current system based on their work, called EffectCheck, presents synonyms that can be used to increase or decrease the level of evoked emotion in each scale. Many other subsequent efforts were less sophisticated, using a mere polar view of sentiment, from positive to negative, such as work by Turney, and Pang who applied different methods for detecting the polarity of product reviews and movie reviews respectively. This work is at the document level. One can also classify a document's polarity on a multi-way scale, which was attempted by Pang and Snyder among others: Pang and Lee expanded the basic task of classifying a movie review as either positive or negative to predict star ratings on either a 3- or a 4-star scale, while Snyder performed an in-depth analysis of restaurant reviews, predicting ratings for various aspects of the given restaurant, such as the food and atmosphere (on a five-star scale). First steps to bringing together various approaches—learning, lexical, knowledge-based, etc.—were taken in the 2004 AAAI Spring Symposium where linguists, computer scientists, and other interested researchers first aligned interests and proposed shared tasks and benchmark data sets for the systematic computational research on affect, appeal, subjectivity, and sentiment in text. Even though in most statistical classification methods, the neutral class is ignored under the assumption that neutral texts lie near the boundary of the binary classifier, several researchers suggest that, as in every polarity problem, three categories must be identified. Moreover, it can be proven that specific classifiers such as the Max Entropy and SVMs can benefit from the introduction of a neutral class and improve the overall accuracy of the classification. There are in principle two ways for operating with a neutral class. Either, the algorithm proceeds by

first identifying the neutral language, filtering it out and then assessing the rest in terms of positive and negative sentiments, or it builds a three-way classification in one step. This second approach often involves estimating a probability distribution over all categories (e.g. naive Bayes classifiers as implemented by the NLTK). Whether and how to use a neutral class depends on the nature of the data: if the data is clearly clustered into neutral, negative and positive language, it makes sense to filter the neutral language out and focus on the polarity between positive and negative sentiments. If, in contrast, the data are mostly neutral with small deviations towards positive and negative affect, this strategy would make it harder to clearly distinguish between the two poles. A different method for determining sentiment is the use of a scaling system whereby words commonly associated with having a negative, neutral, or positive sentiment are given an associated number on a  $-10$  to  $+10$  scale (most negative up to most positive) or simply from 0 to a positive upper limit such as  $+4$ . This makes it possible to adjust the sentiment of a given term relative to its environment (usually on the level of the sentence). When a piece of unstructured text is analyzed using natural language processing, each concept in the specified environment is given a score based on the way sentiment words relate to the concept and its associated score. This allows movement to a more sophisticated understanding of sentiment, because it is now possible to adjust the sentiment value of a concept relative to modifications that may surround it. Words, for example, that intensify, relax or negate the sentiment expressed by the concept can affect its score. Alternatively, texts can be given a positive and negative sentiment strength score if the goal is to determine the sentiment in a text rather than the overall polarity and strength of the text. There are various other types of sentiment analysis, such as aspect-based sentiment analysis, grading sentiment analysis (positive, negative, neutral), multilingual sentiment analysis and detection of emotions.

### 3.

Ethical considerations Issues such as privacy, consent, and bias are crucial since sentiment analysis regularly analyzes personal data without explicit user consent. The potential for misinterpretation and misuse of sentiment data can significantly impact societal norms. Furthermore, the development of ethical frameworks, as seen in projects like SEWA, where Ethical and Industrial Valorisation Advisory Boards are established, is essential for addressing these challenges. These boards help ensure that sentiment analysis technologies are used responsibly, especially in applications involving the recognition of human emotions and behaviors. Such frameworks are vital for guiding the responsible use of sentiment analysis tools, ensuring they promote equity and respect user autonomy, and effectively address both routine and complex ethical issues.